Data Collection –

As per our architecture, Model will identify suspicious activities from video feeds. So the model needs to be trained on activity images. The video file will be converted into image frames and each image frame will be an input to the model. We have trained the model on 4 activities i.e. Punch, Kick, Break and Fall.

Images for each category were collected from internet and saved in folder named as the image category.

Image Augmentation -

As we have very less number of images, we have used below image augmentation techniques.

1. Clockwise rotation
2. Anticlockwise rotation
3. Horizontal Flip
4. Vertical Flip
5. Adding noise to image
6. Blurring the image

Image augmentation is done by using Opencv and numpy.

Image Labeling –

As we are doing image classification, we need to keep the images in the respective folders. We are training our model on four activities. So we have four labels – **Punch, Kick, Fall and Break**.

We have set a threshold of 0.5. If the probability of the maximum class is more than threshold value, then the image will be classified as the predicted value otherwise it will be classified as **Others.**